Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Matrices and Determinants

Learning Objectives

• Define a matrix and identify its order.

• Add, subtract and multiply matrices.

• Find the determinant of 2x2 and 3x3 matrices.

• Find the inverse of a matrix and solve linear equations.

Key Concepts

• Matrix, rows, columns, order (m x n)

• Types: square, row, column, identity, zero matrix

• Addition, subtraction and scalar multiplication

• Matrix multiplication and its conditions

• Determinant, adjoint and inverse

Teaching Methodology (Step by Step)

1. Introduce matrices as a table of marks: rows are students, columns subjects.
2. Practise identifying order and types with quick oral questions.
3. Demonstrate matrix multiplication step by step on the board.
4. Solve a pair of linear equations using the inverse method.

Classroom Activities

• Add and multiply given matrices in pairs; verify each other's work.

• Create a 3x3 matrix from a coded message and find its determinant.

Assessment Questions

1. What is the order of a matrix with 3 rows and 4 columns?
2. When can two matrices be multiplied?
3. Find the determinant of [[2,3],[1,4]].
4. Define an identity matrix.

Homework

• Solve the exercise on matrix multiplication.

• Find inverses of three given 2x2 matrices.

Time Allocation

7 periods of 40 minutes each

Unit 2: Real and Complex Numbers

Learning Objectives

• Classify numbers as natural, whole, integer, rational, irrational.

• Convert recurring decimals to fractions.

• Define complex numbers and perform basic operations.

• Apply laws of exponents and radicals.

Key Concepts

• Number sets: N, W, Z, Q, Q', R

• Rational vs irrational numbers

• Recurring decimals as fractions

• Complex number: a + bi; real and imaginary parts

• Laws of exponents; radicals and radicands

Teaching Methodology (Step by Step)

1. Ask whether 0.333... is rational: spark the recurring-decimal idea.
2. Build the number-system chart from N to R on the board.
3. Introduce i = sqrt(-1) through the puzzle x^2 = -1.
4. Practise exponent laws with rapid-fire questions.

Classroom Activities

• Sort number cards into the correct sets.

• Convert five recurring decimals to fractions in pairs.

Assessment Questions

1. Is 22/7 rational or irrational? Why?
2. Convert 0.666... into a fraction.
3. Simplify (3 + 2i) + (1 - 5i).
4. State any two laws of exponents.

Homework

• Complete the exercise on number classification.

• Simplify ten expressions using exponent laws.

Time Allocation

6 periods of 40 minutes each

Unit 3: Logarithms

Learning Objectives

• Define logarithm and convert between log and exponential forms.

• Apply the laws of logarithms.

• Use common logarithms and antilogarithms from tables.

• Solve logarithmic equations.

Key Concepts

• Logarithm: if a^x = y then log_a(y) = x

• Common log (base 10) and natural log (base e)

• Laws: log(mn), log(m/n), log(m^n)

• Characteristic and mantissa

• Change of base idea

Teaching Methodology (Step by Step)

1. Introduce logs as the 'opposite question' of exponents: 2^? = 8.

2. Prove each law with a numerical example before stating it.

3. Demonstrate reading a log table step by step.

4. Solve logarithmic equations together, then assign similar ones.

Classroom Activities

• Convert ten exponential statements to log form and back.

• Use log laws to expand or condense given expressions.

Assessment Questions

1. Convert 3^4 = 81 into logarithmic form.

2. Simplify log 2 + log 5.

3. Find log(1000) without a calculator.

4. State the three laws of logarithms.

Homework

• Solve the exercise on laws of logarithms.

• Find values of five expressions using the log table.

Time Allocation

6 periods of 40 minutes each

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Unit 4: Algebraic Expressions and Formulae

Learning Objectives

• Identify terms, coefficients and like/unlike terms.

• Apply the basic algebraic formulae (a+b)^2, (a-b)^2, a^2-b^2.

• Expand and simplify algebraic expressions.

• Solve problems using standard identities.

Key Concepts

• Algebraic expression, term, coefficient, variable

• Like and unlike terms

• (a + b)^2 = a^2 + 2ab + b^2

• (a - b)^2 = a^2 - 2ab + b^2

• a^2 - b^2 = (a + b)(a - b)

Teaching Methodology (Step by Step)

1. Show (a+b)^2 geometrically with a square divided into parts.

2. Derive each formula on the board; students copy the steps.

3. Give rapid expansion practice with increasing difficulty.

4. Connect formulas to mental multiplication (e.g. 103^2).

Classroom Activities

• Verify (a+b)^2 with paper squares of chosen side lengths.

• Mental-math race: compute 98^2 using (100-2)^2.

Assessment Questions

1. Expand (x + 5)^2.

2. Factorize x^2 - 49 using a standard identity.

3. What are like terms? Give an example.

4. Find (2a - 3b)^2.

Homework

• Complete the exercise on algebraic identities.

• Write five real-life situations where formulas simplify calculation.

Time Allocation

6 periods of 40 minutes each

Unit 5: Factorization

Learning Objectives

• Factorize expressions by taking common factors.

• Factorize using algebraic identities.

• Factorize trinomials of the form x^2 + px + q.

• Factorize by grouping.

Key Concepts

• Factor and factorization

• Common factor method

• Using identities: a^2 - b^2, (a+b)^2

• Trinomial x^2 + px + q: find two numbers with product q, sum p

• Factorization by grouping in pairs

Teaching Methodology (Step by Step)

1. Present factorization as the reverse of expansion.

2. Demonstrate the common-factor method with simple examples.

3. Teach the trinomial trick: 'product-sum' number search.

4. Give mixed practice so students choose the right method.

Classroom Activities

• Factorize expression cards in teams; fastest correct team wins.

• Create your own factorable trinomial and exchange with a partner.

Assessment Questions

1. Factorize 6x^2 + 9x.

2. Factorize x^2 + 7x + 12.

3. Factorize 4a^2 - 9b^2.

4. What is factorization?

Homework

• Solve the mixed factorization exercise.

• Make a personal 'method chooser' flowchart in the notebook.

Time Allocation

6 periods of 40 minutes each

Unit 6: Algebraic Manipulation

Learning Objectives

• Find the HCF and LCM of algebraic expressions.

• Simplify rational expressions.

• Add and subtract algebraic fractions.

• Simplify expressions using square roots.

Key Concepts

• HCF and LCM of monomials and polynomials

• Rational expression: p(x)/q(x)

• Simplifying by cancelling common factors

• Operations on algebraic fractions

• Square roots of algebraic expressions

Teaching Methodology (Step by Step)

1. Revise numerical HCF/LCM, then move to algebraic versions.
2. Show simplification as 'cancelling the common factor'.
3. Demonstrate addition of fractions with different denominators.
4. Give graded practice from simple to board-level questions.

Classroom Activities

• Find HCF/LCM of expression cards in pairs.

• Simplify rational-expression dominoes: match question to answer.

Assessment Questions

1. Find the HCF of 12x^2 and 18x^3.
2. Simplify (x^2 - 4)/(x + 2).
3. Add: 1/x + 1/y.
4. What is a rational expression?

Homework

• Complete the exercise on HCF and LCM.

• Simplify ten given rational expressions.

Time Allocation

6 periods of 40 minutes each

Unit 7: Linear Equations and Inequalities

Learning Objectives

• Solve linear equations in one variable.

• Solve simultaneous linear equations by elimination and substitution.

• Solve linear inequalities and represent solutions on a number line.

• Solve word problems using linear equations.

Key Concepts

• Linear equation in one and two variables

• Elimination method

• Substitution method

• Inequality symbols and their rules (sign flip on multiplying by negative)

• Solution set on the number line

Teaching Methodology (Step by Step)

1. Translate a shopping word problem into an equation together.
2. Demonstrate elimination: 'add to kill one variable'.
3. Show substitution with a simple two-variable example.
4. Practise inequalities, stressing the sign-flip rule.

Classroom Activities

• Solve equation cards; match each equation to its solution.

• Write a word problem for a given equation and exchange.

Assessment Questions

1. Solve: $3x + 5 = 20$.
2. Solve by elimination: $x + y = 7$, $x - y = 3$.
3. Solve and graph: $2x - 3 < 7$.
4. When does the inequality sign reverse?

Homework

• Solve the word-problem exercise.

• Solve and graph five inequalities.

Time Allocation

6 periods of 40 minutes each

Unit 8: Congruent Triangles

Learning Objectives

• Define congruent triangles and their corresponding parts.

• Prove triangles congruent using SSS, SAS, ASA postulates.

• Prove the isosceles triangle theorem.

• Apply congruence in simple riders.

Key Concepts

• Congruence: same shape and size

• Corresponding vertices, sides and angles

• SSS, SAS, ASA, RHS criteria

• Isosceles triangle theorem and its converse

• Riders: short deductive proofs

Teaching Methodology (Step by Step)

1. Cut two identical paper triangles: show they overlap exactly.
2. Test each criterion by constructing triangles with given data.
3. Prove the isosceles theorem step by step on the board.
4. Guide students through two riders before independent work.

Classroom Activities

• Construct triangles from given measurements and check congruence.

• Prove in pairs: base angles of an isosceles triangle are equal.

Assessment Questions

1. State the SSS postulate.
2. What does SAS stand for?
3. In triangle ABC, AB = AC. What can you say about angle B and C?
4. Name the four congruence criteria.

Homework

• Write proofs of the congruence postulates in the notebook.

• Solve three given riders.

Time Allocation

6 periods of 40 minutes each